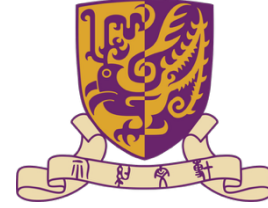


Data Analytics for Personalized Genomics and Precision Medicine

L19: Deep Learning and Biomedical Imaging



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Main Points

1. T-SNE
2. Artificial intelligence VS Machine learning VS Deep learning
3. Deep Neural Networks

T-SNE (Review)

As a review of last class, T-SNE (T-Distributed Stochastic Neighbor Embedding) is a dimension reduction tool, which can be used instead of PCA. It is especially good for embedding high-dimensional data for visualization in a low-dimensional space of 2 or 3 dimensions.

Similar objects are modelled by nearby points and different objects are modelled by distant points with high probability in an iterative process. The initialization is usually random.

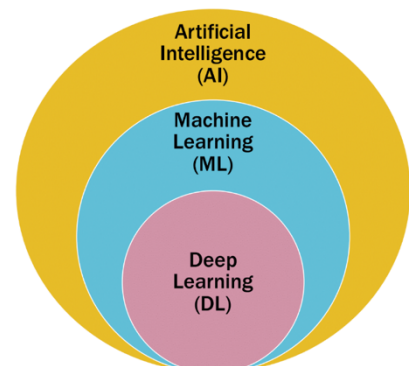
Artificial intelligence VS Machine learning VS Deep learning

Although often used interchangeably, these 3 concepts should not be confused, and their differences should remain clear.

Artificial intelligence is a general concept referring to all techniques which permit computers to mimic human behavior or acts. For example, a robotic arm on a fabric.

Machine learning is a more advanced type of artificial intelligence, when machines perform a task without using explicit instructions, using patterns and insights from the data. For example, a self-driving car.

Finally, deep learning is a subset of Machine Learning, it refers to a family of algorithms which take advantage of multi-layer neural networks. For example, ChatGPT.



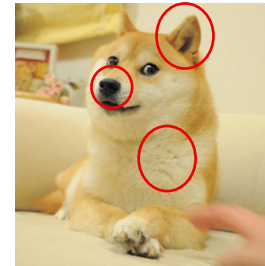
Deep Learning for Disease Screening

Deep learning can be especially good at classifying multi-dimensional data, including images. The process is the usual for training machine learning models. Training data with labels is used to fit the model, then the data to be classified is passed as input and the model returns a prediction label.

Before, we had seen in class how to use logistic regression to do this task. However, logistic regression presented some problems, like that the relation between outputs and inputs may be nonlinear or very complex, being incompetent to classify this kind of relations.

Nevertheless, Deep Neural Networks can solve complicated problems by simply increasing the number of nodes and layers and adding non-linear functions. One general way to do it is by using fully connected layers, which is able to approximate any relation if enough nodes and layers are used, using the Universal Approximation Theorem.

For example, when classifying an image as a dog, people would usually unconsciously analyze some particular features common on dogs, such as their hoop, their nose and their ears. Deep Neural Networks can also learn those underlying characteristics by using several nodes and layers.



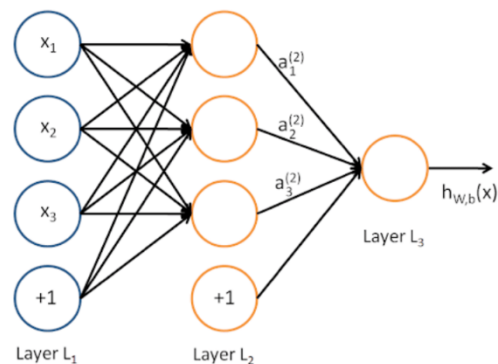
Feature Extraction

New features are extracted using linear or non-linear combinations of the original features. These new features may not have physical interpretation or meaning, but they can help the model to properly predict the classes.

Number of Parameters

The number of parameters on a dense layer is equivalent to the number of connections of each node from the first layer to the second one, including the bias nodes. In the next graph, it can be seen like the number of lines connecting one layer with the next one ($4 \times 3 + 4 = 16$).

If we create a model with dozens and nodes and layers, parameters will surely be on the order of thousands or millions, resulting in a very complicated function, which comes in handy for resolving complex problems with a huge amount of data.



Problems of Fully Connected Networks

Determining the right number of nodes and layers can be time consuming, the storage and running time can easily get very large and a very complex model can lead to overfitting.

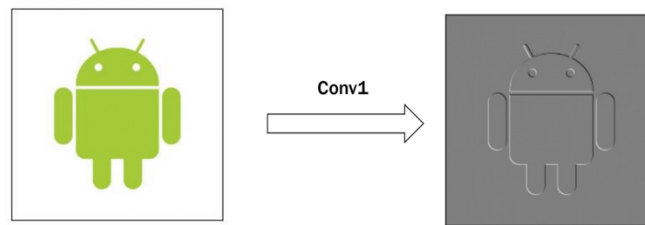
Bessies, if we wish to classify images, Full Connected Networks will not be able to understand the special information of the input.

Convolutional Layers

Convolutional layers can help us in this case, since these are able to capture the patch information, no matter where it is (translation invariance), and focusing on local regions before getting the whole picture.

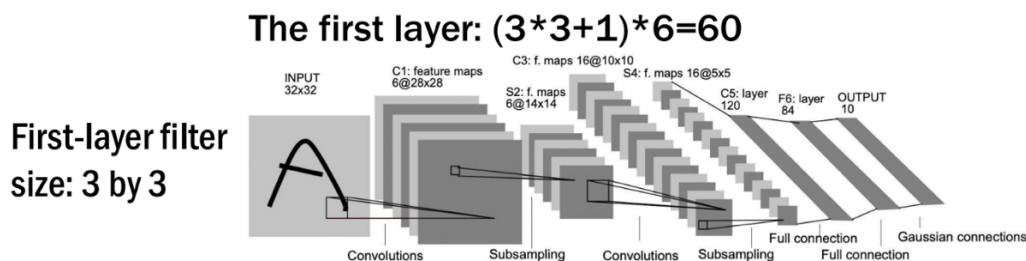
How they work

Convolutional layers work by applying convolutional operations to the data, in other word, calculating the product of the input and a kernel or filter. The filter reduces the data, leaving only certain features, hence, alleviating the overfitting issue and detecting translation invariant features.



Number of Parameters

To get the number of parameters we need to multiply the dimensions of the filter and the number of channels of the next layer. The bias must also be considered.



Padding

Padding is a tool used to alleviate the dimension reduction of the output. It consists of adding n borders filled with 0 to the input data.

Input

0	0	0	0	0
0	0	1	2	0
0	3	4	5	0
0	6	7	8	0
0	0	0	0	0

Stride

Stride is a tool that permits to make data initially far, close in a smaller matrix. It can be understood as the number of rows/columns that the filter will move in each step.

Input

0	0	0	0	0
0	0	1	2	0
0	3	4	5	0
0	6	7	8	0
0	0	0	0	0

Pooling

Pooling is a special filter which summarizes regions of data by returning their maximum value or their average. It can also be combined with padding and stride.

